Q_{10} calculator by condition

Temperature dependence of the circadian period, fitted and compared across conditions

User manual

Web app: https://dxda6216.github.io/q10_calculator_by_condition/index.html

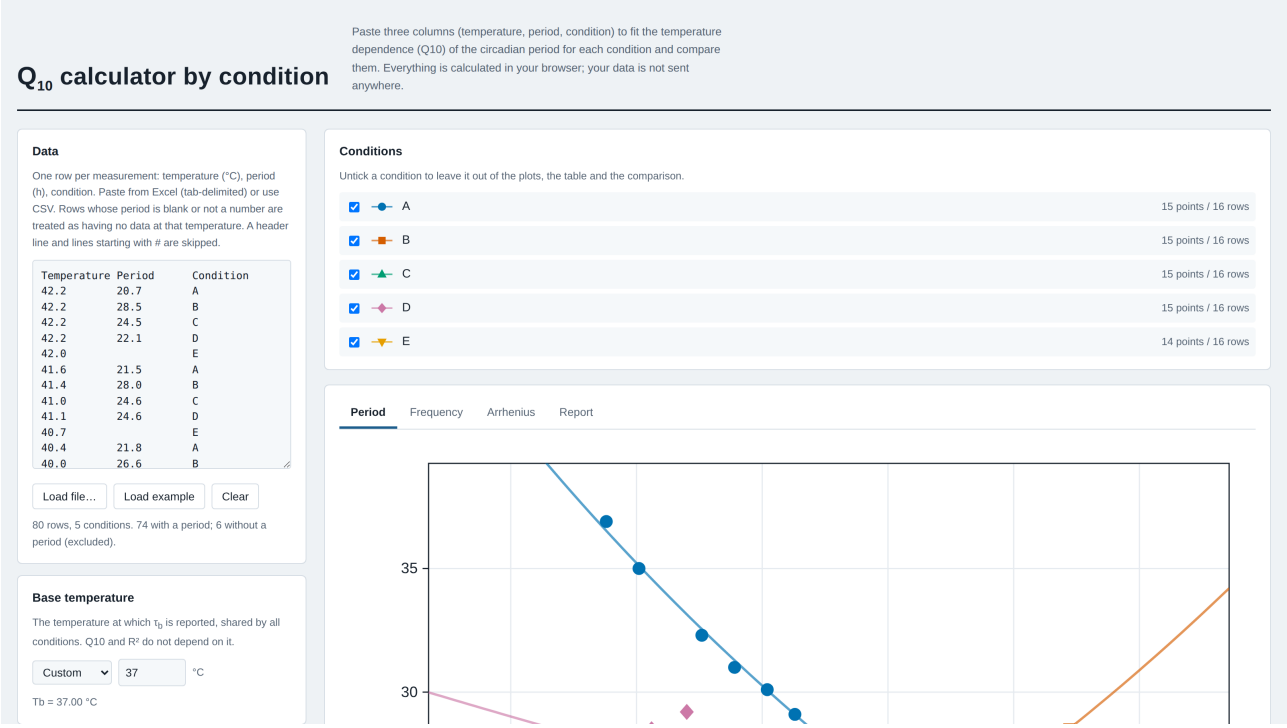


Figure 1. The app with the built-in example data loaded. Settings are in the left column; conditions, plots, results and the comparison are on the right.

1. Overview

The app takes a list of temperatures, circadian periods and a label for each measurement (cell type, drug treatment, medium, genotype and so on). For each label (a “condition”) it fits the exponential Q_{10} model to period against temperature, reports Q_{10} with its standard error and 95% confidence interval, and plots all conditions together on period, frequency and Arrhenius plots. It then tests whether Q_{10} differs between conditions. Optionally, a polynomial of period on temperature is fitted to each condition.

It follows the fitting method of the single-condition Q10 calculator for circadian period and applies it to each condition separately.

- **Runs entirely in your browser.** Your data are never uploaded. The page also works offline once loaded, and the single file index.html can be saved and opened locally.
- **Updates live.** Every change to the data or settings recalculates immediately; there is no Calculate button. Ctrl+Enter (⌘+Enter on a Mac) forces a recalculation.
- **Works in current desktop browsers** (Chrome, Edge, Firefox, Safari). Tablets and phones work too, but pasting data is easiest on a computer.

2. Quick start

1. Open the app. Example data with five conditions (A–E) are already loaded, so you can explore every feature straight away.
2. Select three columns in Excel (temperature, period, condition), copy, click in the **Data** box, select all and paste.
3. Check the line under the box: it reports how many rows, conditions and valid periods were read.
4. Choose the **Base temperature** (default 37 °C) and, if needed, restrict the **Temperature range for the fit**.
5. Read Q_{10} for each condition in **Results by condition** and whether they differ in **Comparison between conditions**.
6. Save the figure (PNG or SVG), the results table (CSV) or the full report (TXT).

3. Input data

Enter one measurement per row with three columns, in this order:

Column	Content	Notes
1	Temperature (°C)	Must be a number. A row whose temperature is not a number is skipped (the first such row is taken as the header).
2	Period (h)	A positive number. If the cell is blank or not a number (e.g. “arrhythmic”, “n.d.”), the row is kept for counting but treated as having no data at that temperature.
3	Condition	Any text, e.g. “HeLa”, “10 µM drug X”, “1 mM glucose + 2 mM GlutaMAX”. Rows with identical text form one condition. An empty cell gives the condition “(no label)”.

Rows from different conditions can be mixed in any order. Conditions are listed, coloured and given marker shapes in the order in which they first appear. A header line and lines beginning with # are ignored.

Example (tab-delimited; the second row has no period and is excluded):

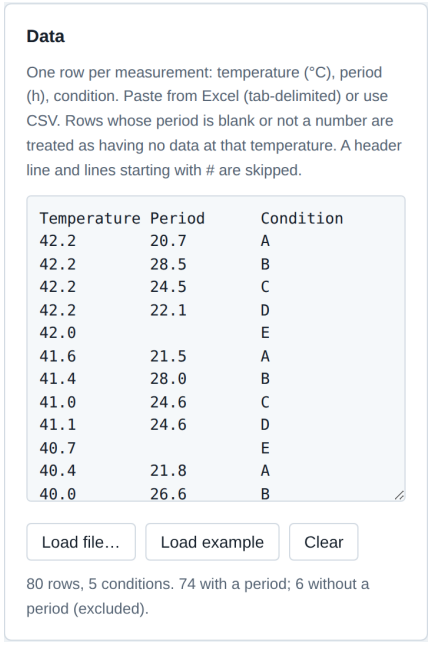
```
Temperature   Period        Condition
42.2          20.7          Control
42.0                        10 µM drug X
40.4          21.8          Control
40.0          26.6          10 µM drug X
```

Delimiters

- **Tab** (recommended): what you get when you copy cells from Excel, Numbers or Google Sheets. Condition names may then contain spaces and commas.
- **Comma or semicolon** (CSV files). Put a condition name in double quotes if it contains a comma, e.g. `37.0,24.5,"glucose, 1 mM"`.
- **Spaces only**: avoid. With space-separated values a blank period cannot be told apart from the condition name.

Loading and clearing

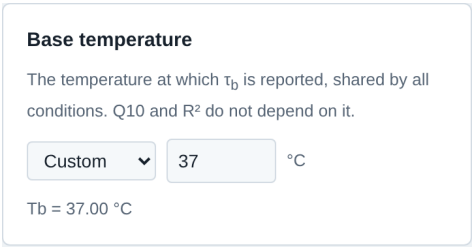
- **Load file…** reads a .csv, .tsv or .txt file from your computer into the box.
- **Load example** restores the built-in example data (Section 11).
- **Clear** empties the box.
- The line under the buttons reports the number of rows and conditions read, how many rows have a valid period, and how many were excluded.



4. Settings

4.1 Base temperature

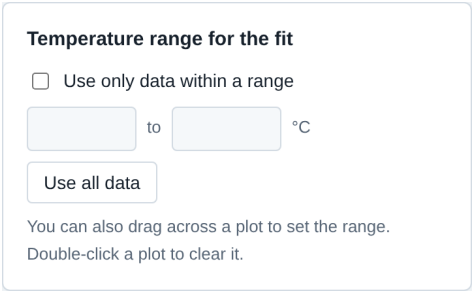
τ_b, the period predicted by the fit at the base temperature T_b, is reported for every condition. T_b is shared by all conditions so that their τ_b values can be compared directly. Choose **Custom** (default, 37 °C) and type a value, or let the app use the **Minimum**, **Maximum**, **Average** or **Median** of all temperatures used in the fit (pooled over the conditions that are switched on). The value in use is shown below the selector. Q_{10}, its SE and R^2 do not depend on T_b.



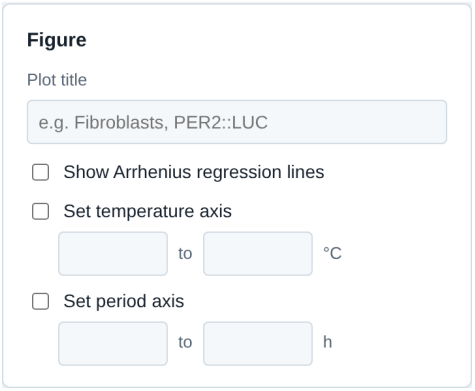
4.2 Temperature range for the fit

Tick **Use only data within a range** and enter the low and high limits (inclusive) to fit Q_{10} only to data within that range, for example to exclude temperatures at which the clock becomes unstable. The same range applies to every condition. Either limit may be left empty to leave that side open.

- You can also **drag horizontally across any plot** to set the range; the limits are filled in for you.
- **Double-click a plot** or press **Use all data** to remove the range.
- The range is shaded on the plots. Points inside it are filled, points outside are drawn open.

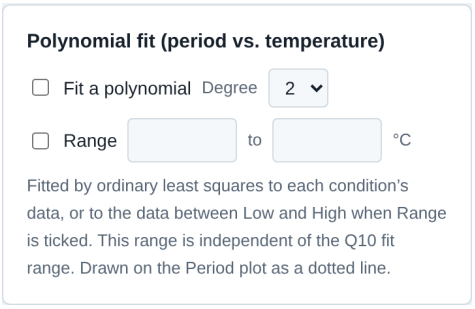


4.3 Figure



- **Plot title**: optional text printed above each plot.
- **Show Arrhenius regression lines**: adds the straight-line regression of ln(frequency) on 1000/T to the Arrhenius plot as a dashed line in each condition's colour.
- **Set temperature axis / Set period axis**: fix the axis limits instead of using automatic scaling. The frequency and Arrhenius axes are derived from these limits.
- Automatic scaling: the temperature axis runs from the lowest temperature minus one third of the temperature span to the highest temperature plus one third of the span, as in the single-condition calculator. The period axis is fitted to the data with a small margin.
- The plotting area always has a width-to-height ratio of 4:3.

4.4 Polynomial fit (period vs. temperature)



Tick **Fit a polynomial** and choose the degree (1–6). Tick **Range** and enter limits to fit only the data between them. This range is independent of the Q_{10} fit range. The polynomial does not affect any Q_{10} result; it is a descriptive curve for data that do not follow the exponential model (for example, a U-shaped dependence). It is drawn on the Period plot only.

4.5 Report

Two options add tab-delimited blocks to the end of the report, ready to paste into a spreadsheet or graphing program:

- **Include the data in the report**: the data used, with a column showing whether each point was inside the Q_{10} fit range (1) or not (0).
- **Include fitted curves**: for each condition, 200 points of the fitted Q_{10} curve (Temperature, Period, Frequency, In_fit_range) over the plot's temperature axis, and, when the polynomial fit is on, 200 points of the polynomial curve (Temperature, Period) over the condition's temperature span plus 10% on each side.

5. Conditions

All conditions found in the data are listed with their colour and marker, the number of points with a valid period, and the number of rows. Untick a condition to remove it from the plots, the results table, the comparison and the exported files. Colours come from a colour-blind-safe palette, and each condition also has its own marker shape so that figures remain readable in greyscale.



Figure 3. The Conditions panel.

6. Plots

Use the tabs above the plot to switch between three views. Every view shows all active conditions together, with a legend under the plot that lists Q_{10} ± SE and n for each condition.

Tab	x axis	y axis
Period	Temperature (°C)	Period (h)
Frequency	Temperature (°C)	Frequency = 24 / period (day^{-1})
Arrhenius	1000 / T (K^{-1}), T in kelvin	ln(frequency)

Line and marker styles

- **Filled markers**: data used in the Q_{10} fit. **Open markers**: data outside the fit range.
- **Solid line**: the fitted Q_{10} model, drawn across the whole temperature axis. It is darker inside the fit range and lighter outside it, so the extrapolated part is easy to see.
- **Dotted line** (Period plot): polynomial fit, drawn over the condition’s temperature span plus 10% on each side.
- **Dashed line** (Arrhenius plot, optional): Arrhenius regression line.
- Fit lines are drawn slightly translucent so that the data points stand out.

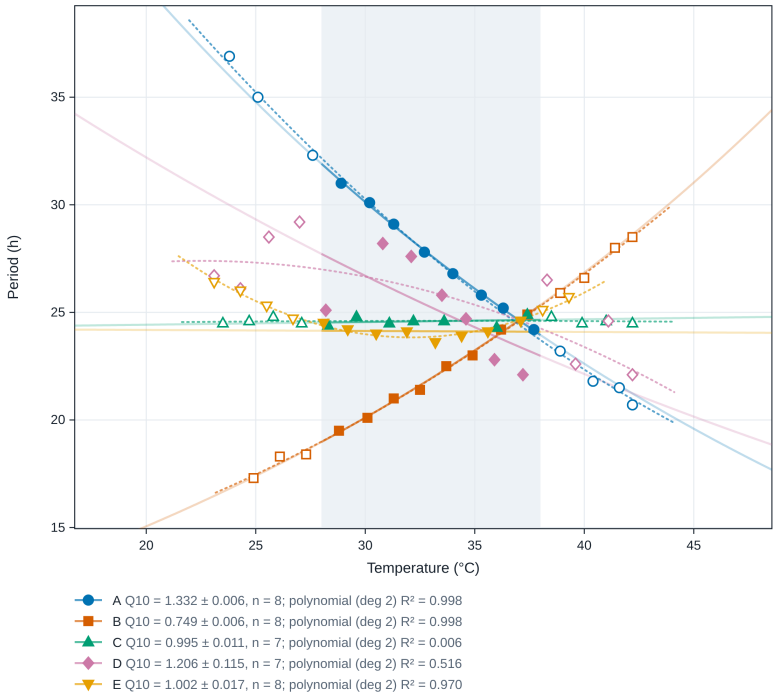


Figure 4. Period plot with the Q_{10} fit restricted to 28–38 °C (shaded) and a second-degree polynomial fit switched on. Condition E is U-shaped: the exponential model gives $Q_{10} \approx 1$ with $R^2 \approx 0$, while the polynomial describes it well.

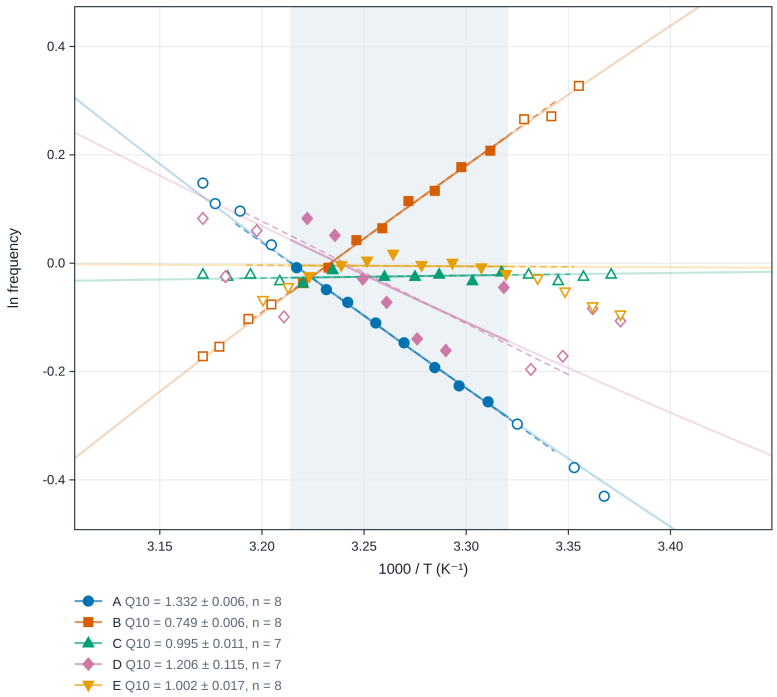


Figure 5. Arrhenius plot with the regression lines switched on.

7. Results by condition

Results by condition

Condition	n	Q10 ± SE	95% CI	0.67 Q10 1.61	τb ± SE (h)	R²	Ea (from Q10) kJ/mol	Ea (regression) kJ/mol
● A	8	**1.332** ± 0.006	1.318–1.347		24.65 ± 0.05	0.999	22.40	22.37 ± 0.32
■ B	8	**0.749** ± 0.006	0.733–0.765		24.62 ± 0.09	0.995	-22.55	-22.49 ± 0.69
▲ C	7	**0.995** ± 0.011	0.966–1.024		24.64 ± 0.15	0.039	-0.40	-0.39 ± 0.89
◆ D	7	**1.206** ± 0.115	0.944–1.541		23.42 ± 1.13	0.445	14.57	15.70 ± 7.36
▼ E	8	**1.002** ± 0.017	0.961–1.045		24.10 ± 0.22	0.002	0.15	0.18 ± 1.33

τb is the period at Tb = 37.00 °C. The dotted line marks Q10 = 1 (perfect temperature compensation).

Polynomial fits, degree 2 (all data)

- **A**: $\tau = 70.0782 - 1.72827\ T + 0.0133733\ T^2$
  n = 15, R² = 0.9981, adjusted R² = 0.9978, residual SE = 0.233 h
- **B**: $\tau = 11.3379 + 0.0104250\ T + 0.00939223\ T^2$
  n = 15, R² = 0.9981, adjusted R² = 0.9978, residual SE = 0.169 h
- **C**: $\tau = 24.1636 + 0.0261716\ T - 0.000385577\ T^2$
  n = 15, R² = 0.0058, adjusted R² = -0.1599, residual SE = 0.180 h
- **D**: $\tau = 20.7970 + 0.586760\ T - 0.0130466\ T^2$
  n = 15, R² = 0.5163, adjusted R² = 0.4356, residual SE = 1.770 h
- **E**: $\tau = 58.2842 - 2.14559\ T + 0.0334106\ T^2$
  n = 14, R² = 0.9699, adjusted R² = 0.9644, residual SE = 0.161 h

Figure 6. Results table and, below it, the polynomial fit summary (shown when the polynomial fit is on).

Column	Meaning
n	Number of points used in the Q_{10} fit (within the fit range, with a valid period).
Q_{10} ± SE	Fitted Q_{10} and its standard error. $Q_{10} > 1$: the period shortens as temperature rises; $Q_{10} = 1$: perfect temperature compensation; $Q_{10} < 1$: overcompensation (the period lengthens as temperature rises).
95% CI	Confidence interval of Q_{10}, computed on the log scale and therefore asymmetric. Needs n ≥ 3.
Forest plot	Q_{10} (marker) and its 95% CI (bar) for every condition on a common axis. The dotted line marks $Q_{10} = 1$. A CI that does not cross the line indicates a significant temperature dependence.
τ_b ± SE (h)	Fitted period at the base temperature T_b.
R^2	Coefficient of determination of the Q_{10} fit on the period scale. A low R^2 can mean either that the period hardly changes with temperature ($Q_{10} \approx 1$, e.g. condition C) or that the data are noisy or not exponential (D, E). Read it together with the CI.
Ea (from Q_{10})	Apparent activation energy from Q_{10} (Section 10.4), kJ/mol.
Ea (regression)	Activation energy ± SE from the Arrhenius regression, kJ/mol.

A condition with fewer than two points, or with all points at one temperature, is shown as “Not fitted” with the reason. With exactly two points, Q_{10} is calculated but no SE or CI.

8. Comparison between conditions

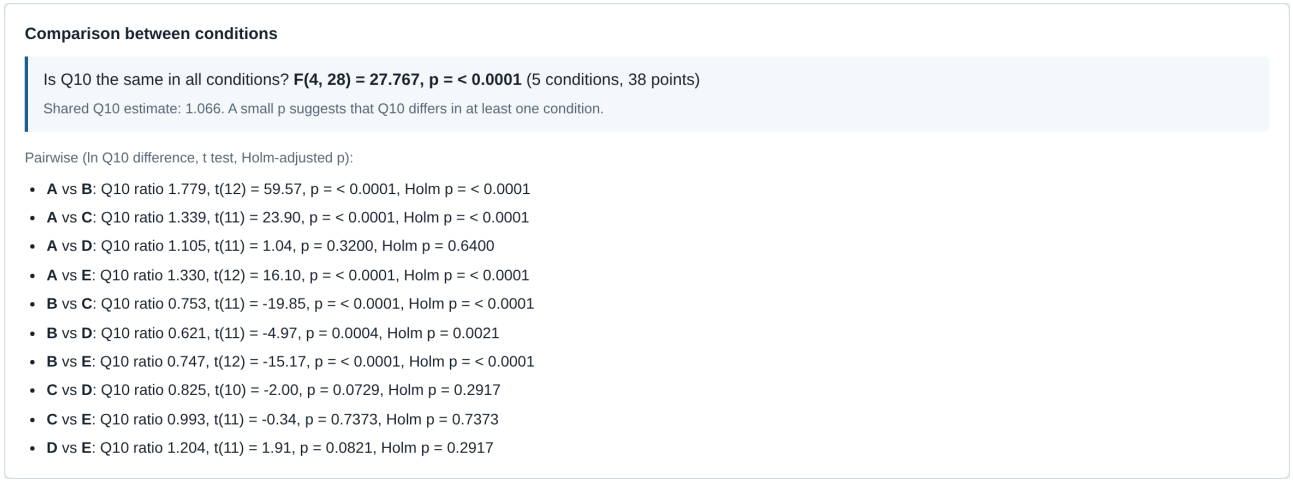


Figure 7. Comparison of Q_{10} between the example conditions.

Overall test. Is Q_{10} the same in all conditions? The app answers this with an extra sum-of-squares F test (details in Section 10.5). It also reports the shared Q_{10} estimated under the null hypothesis. A small p value means that Q_{10} differs in at least one condition.

Pairwise comparisons. Every pair of conditions is listed with the ratio of their Q_{10} values, a t statistic, its p value and the Holm-adjusted p value for multiple comparisons. Use the Holm-adjusted value when you look at several pairs.

Only conditions with at least three points enter the comparison, and at least two such conditions are needed. Both tests treat the fitted points as independent. If your periods are averages of replicates, or you have replicate cultures at each temperature, consider whether that assumption holds; the tests are best used as a guide alongside the confidence intervals.

9. Report and saving

The **Report** tab shows a plain-text summary of everything: settings, the results for each condition (including the Arrhenius regression equation), the polynomial fits, the comparison, and the results table in tab-delimited form, ready to paste into a spreadsheet.

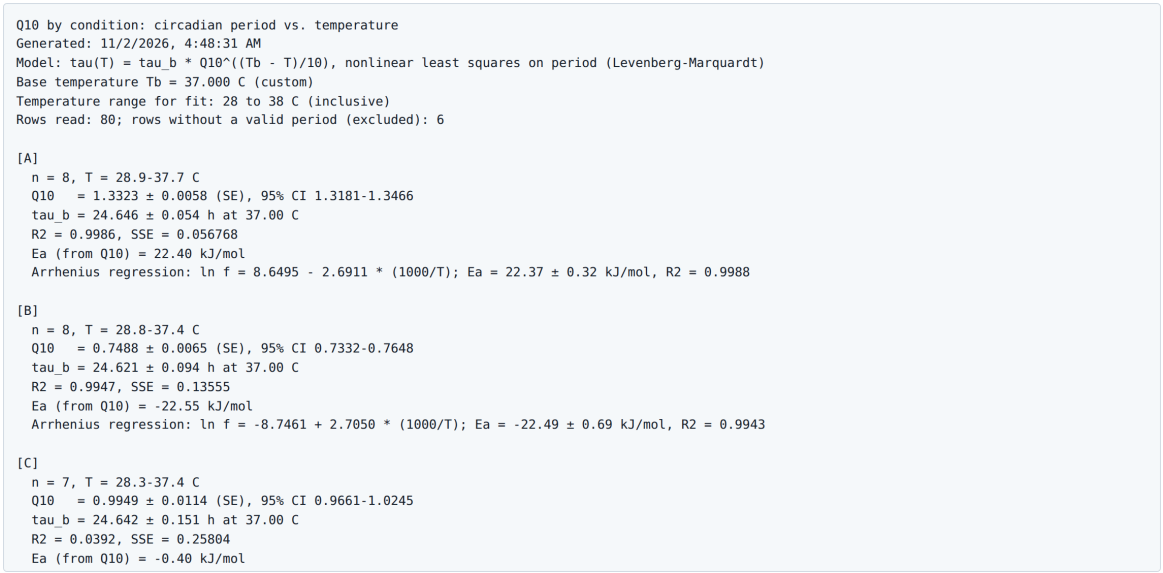


Figure 8. The Report tab (top of the report).

Button	Output	File name
Save figure (PNG)	Period, Frequency and Arrhenius plots stacked in one image, at twice screen resolution, always on a white background.	q10_by_condition_YYYY-MM-DD.png
Save figure (SVG)	The same figure as a vector file that can be edited in Illustrator, Inkscape or PowerPoint.	q10_by_condition_YYYY-MM-DD.svg
Save results (CSV)	One row per condition with all fitted values (and polynomial R^2, adjusted R^2 and residual SE when the polynomial fit is on).	q10_summary_YYYY-MM-DD.csv
Save report (TXT)	The full text report.	q10_report_YYYY-MM-DD.txt
Copy report	Copies the report to the clipboard.	–

Saved files contain only the conditions that are switched on and use the current settings.

10. Methods

10.1 Q_{10} model and fit

For each condition, period τ at temperature T is modelled as

$$\tau(T) = \tau_b \cdot Q_{10}^{(T_b - T)/10}$$

where T_b is the base temperature and τ_b the period at T_b. The parameters ln τ_b and ln Q_{10} are estimated by non-linear least squares on the period itself (not on log period), using the Levenberg–Marquardt algorithm. Starting values come from a straight-line regression of ln τ on T. Because the model is fitted on the period scale, the residuals are in hours and $R^2 = 1 - SSE / SST$ refers to the period.

10.2 Standard errors and confidence intervals

The covariance matrix of (τ_b, Q_{10}) is $s^2(J^TJ)^{-1}$, where J is the Jacobian of the model with respect to τ_b and Q_{10} at the solution and $s^2 = SSE / (n - 2)$. The standard errors are the square roots of its diagonal. The 95% CI of Q_{10} is

$$\exp(\ln Q_{10} \pm t_{0.975,\, n-2} \cdot \mathrm{SE}_{Q10} / Q_{10}\ )$$

i.e. it is computed on the log scale (SE of ln Q_{10} by the delta method) and back-transformed, so it stays positive and is asymmetric. Standard errors require n ≥ 3.

10.3 Frequency and Arrhenius regression

Frequency f = 24 / τ (cycles per day). For the Arrhenius regression, ln f is regressed on 1000 / T_K (T_K = T + 273.15) by ordinary least squares, and

$$E_a = -\text{slope} \times 1000 \times R, \quad R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$$

reported in kJ/mol with the SE derived from the SE of the slope.

10.4 Activation energy from Q_{10}

$$E_a \approx R \cdot T_1 \cdot T_2 \cdot \ln Q_{10} / 10$$

with T_1 and T_2 the lowest and highest temperatures used in the fit, in kelvin. A negative Ea corresponds to $Q_{10} < 1$.

10.5 Overall F test

Two models are fitted to all active conditions with n ≥ 3 together (k conditions, N points in total). The **separate** model gives each condition its own τ_b and Q_{10} (2k parameters; its SSE is the sum of the individual fits). The **common** model gives each condition its own τ_b but a single shared Q_{10} (k + 1 parameters). Then

$$F = [(\mathrm{SSE}_{\mathrm{common}} - \mathrm{SSE}_{\mathrm{separate}}) / (k - 1)] / [\mathrm{SSE}_{\mathrm{separate}} / (N - 2k)]$$

is compared with the F distribution with k − 1 and N − 2k degrees of freedom. The test assumes similar residual scatter in all conditions.

10.6 Pairwise tests

$$t = (\ln Q_{10,A} - \ln Q_{10,B}) / \sqrt{(\mathrm{SE}_A^{\,2} + \mathrm{SE}_B^{\,2})}, \quad \mathrm{df} = n_A + n_B - 4$$

where SE is the standard error of ln Q_{10}. Two-sided p values are adjusted with Holm’s step-down method. This is an approximate (Wald-type) test.

10.7 Polynomial fit

$\tau = a_0 + a_1 T + \ldots + a_d T^d$ is fitted by ordinary least squares. For numerical stability the calculation uses centred and scaled temperature internally, and the coefficients are converted back to raw temperature (°C) for reporting, with their standard errors propagated accordingly. A fit needs at least d + 1 different temperatures; standard errors, adjusted R^2 and the residual standard error need more points than coefficients ($df = n - d - 1 > 0$). The report gives coefficients to 10 significant digits, because high-degree coefficients in raw temperature are large and cancel each other.

11. Example data

The built-in example contains simulated (not experimental) data: five conditions, A–E, each with 16 rows between about 23 and 42 °C and a few blank periods. They were designed to show typical situations:

Condition	Simulated behaviour	Fitted Q_{10} (all data, T_b = 37 °C)
A	Exponential, Q_{10} = 1.35, small noise	1.354 (95% CI 1.344–1.365), R^2 = 0.998
B	Exponential, Q_{10} = 0.75 (overcompensation)	0.751 (0.746–0.757), R^2 = 0.998
C	Constant period (perfect compensation)	1.000 (0.993–1.006), $R^2 \approx 0$
D	Q_{10} = 1.1 with large scatter (±2.5 h)	1.107 (1.038–1.180), R^2 = 0.48
E	U-shaped (quadratic); no period at the two highest temperatures	1.025 (0.987–1.065), R^2 = 0.14; a 2nd-degree polynomial gives R^2 = 0.97

12. Tips and troubleshooting

- **A condition is split in two.** The labels differ, often by a trailing space, a full-width character or different capitalisation. Check the list in the Conditions panel.
- **Fewer rows than expected.** The status line reports rows whose temperature is not a number. Check for units typed into the cells (e.g. “37 °C”) or decimal commas (use 37.5, not 37,5).
- **Periods are misread with space-separated data.** Paste tab-delimited data instead (Section 3).
- **Q_{10} has a very wide CI.** The temperature range is narrow or the data are noisy. Extending the range, or adding temperatures, helps more than adding replicates at the same temperatures.
- **The model fits poorly.** Check the Period plot: if the dependence is curved (e.g. condition E), consider a restricted fit range or report the polynomial fit as well.

13. Reference

Q10 calculator by condition: https://dxda6216.github.io/q10_calculator_by_condition/index.html

Single-condition version: https://dxda6216.github.io/q10/index.html